

AASTHA SURI

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SUMMARY

Data Science graduate with a strong foundation in Computer Science, skilled in Python and R. Experienced in leveraging machine learning and advanced analytics for extracting actionable insights from large datasets, driving data-informed decision-making.

EDUCATION

Rutgers University

Master of Science in Data Science

New Brunswick, NJ, USA

2023-2025

Relevant Coursework: Data Mining, Data Wrangling, Data Structures & Algorithms, Probability & Inference, Regression Time Series, Data Management (SQL, NoSQL, MongoDB), Algorithmic Trading and Portfolio Management (ATPM), Foundations of Financial Statistics and Risk Management (FSRM), Statistical Learning, Statistical Modeling and Computing.

Bharati Vidyapeeth's College of Engineering

B. Tech, Computer Science and Engineering, 3.71

New Delhi, India

2019-2023

PROFESSIONAL EXPERIENCE

Publicis Resources (Publicis Groupe)

Intern Technology

Gurugram, India

08/2022-09/2022

- Proficient in utilizing Microsoft Azure for cloud-based solutions, streamlining data processing tasks.
- Skilled in leveraging Microsoft Azure for Machine Learning and Data Analysis tasks, contributing to data-driven decision-making.
- Employed Microsoft Azure for elementary Machine Learning and Data Analysis tasks.

Aspexx Health Solutions Private Limited (CUREYA)

Technical Intern

Noida, India

07/2022-09/2022

- Researched machine learning and statistical modeling techniques to enhance domain expertise in predictive analytics.
- Implemented the Prophet Algorithm for time-series forecasting, accurately predicting Monkeypox cases over five days.
- Executed data preprocessing, model development, and performance evaluation to improve predictive accuracy.

PROJECTS

Option Pricing with Machine Learning

- Situation: Black-Scholes struggles with real-world option pricing, especially for volatile stocks.
- Task: Improve pricing accuracy using machine learning.
- Action: Analyzed 13 tickers, engineered features using Greeks, and trained Random Forest & XGBoost models on residual errors.
- Result: Reduced MAE by 33%, improved R^2 from 0.68 to 0.92, and enhanced pricing accuracy for complex market conditions.

Trading Strategy Based on Economic Indicators

- Situation: Developed a trading strategy to predict stock market movements using economic indicators (GDP, unemployment, inflation, and consumer confidence).
- Task: Created a predictive model combining machine learning and economic data to optimize trading decisions using Python.
- Action: Collected and processed economic data, built machine learning models (Random Forest, Gradient Boosting), and developed a backtesting framework with risk management (stop-loss, take-profit).
- Result: Achieved a \$15,000 portfolio value with strong risk-adjusted returns, high Sharpe ratios, and low drawdowns, demonstrating effective strategy performance.

TECHNICAL AND OTHER SKILLS

- Programming Languages:** Python, R, C/C++
- Data Science & Machine Learning:** NumPy, Pandas, Scikit-Learn.
- Data Analysis & Visualization:** Advanced Statistical Techniques, Tableau, Power BI
- Tools & Platforms:** Microsoft Office Suite, Microsoft Azure, SQL

RESEARCH PUBLICATIONS

- An Ensemble Learning Approach for Classifying Illicit Transactions in Bitcoin – 2023 3rd International Conference on Technological Advancements in Computational Sciences (ICTACS), Jul 2023
- A Hierarchical Clustering Approach for the Identification of Molecular Subtypes in Colon Cancer Using Gene Expression Data – Intelligent Medicine (Elsevier B.V.), Nov 2022
- Prediction and Analysis of Polycystic Ovary Syndrome (PCOS) Using Machine Learning – Computational Intelligence for Engineering and Management Applications (CIEMA 2022), Aug 2021

LANGUAGE SKILLS

- English (Full Professional Proficiency)
- Hindi (Native/Bilingual Proficiency)